Iowa State University
Dependable Data Driven Discovery

EMNLP2026 BUDAPEST

Paper

Bridging Lexical Divergence: LLM-Assisted, Cost-Efficient, Zero-Shot Scientific Entity Linking

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Summary

THE PURPOSE

Link textual mention from scientific paper even when mention and referent entity has no shared words.

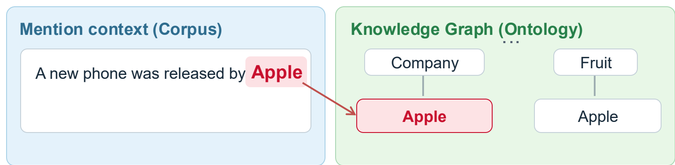
Sci-ZSEL creates reusable training supervision from unlabeled corpus, without human annotation.

1. **New animal science benchmark**
   PubMed mentions linked to three livestock trait ontologies.
2. **Sci-ZSEL method**
   Select subset of entities, filtered LLM-generated alias names, construct pseudo-label then train
3. **Evidence across five datasets**
   Improve the backbones; **Synonym + Sci-ZSEL alias** wins in most settings, especially when no words overlap.

Problem

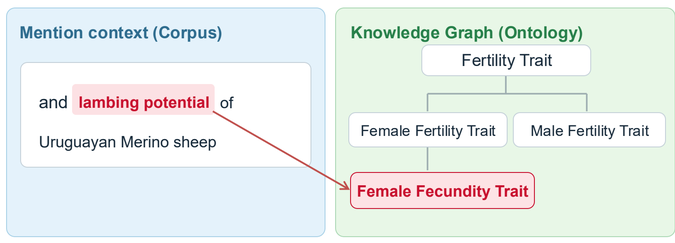
TWO DIFFERENT MATCHING PROBLEMS

Entity Linking in General Domain



Several candidates are named "Apple."

Entity Linking in Scientific Domain



The mention and correct referent share no words.

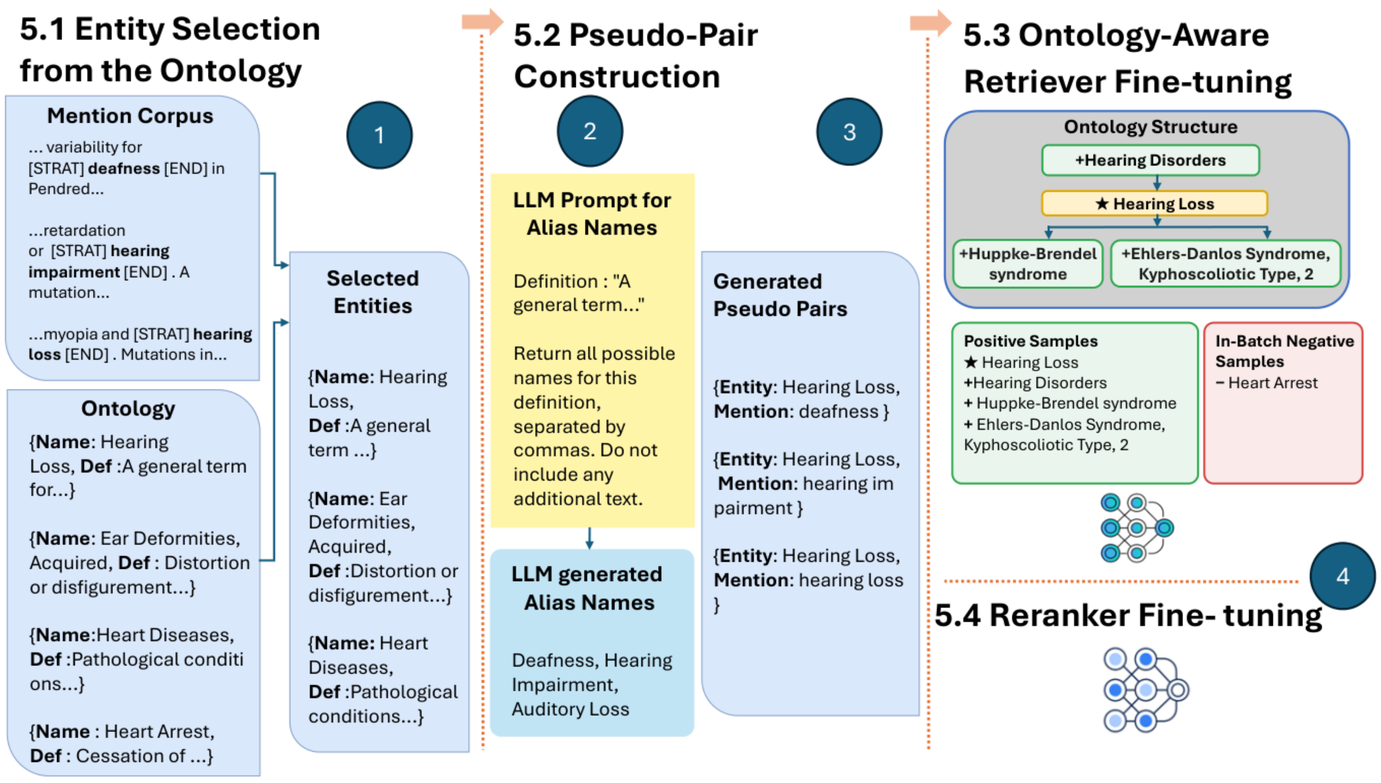
WHY EXISTING APPROACHES STRUGGLE

- Specialist terms
- Ignored hierarchy
- Annotation cost
- Token cost
- Non-reusable generated supervision

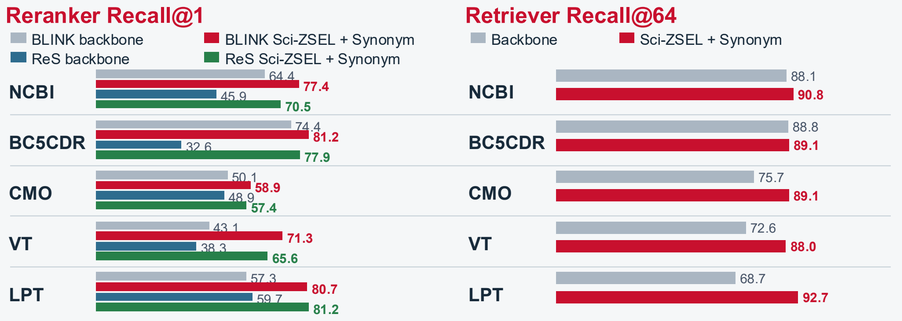
Solution

HOW SCI-ZSEL WORKS

The overview shows how unlabeled mentions become pseudo-pair training signal for the Entity Linking system.

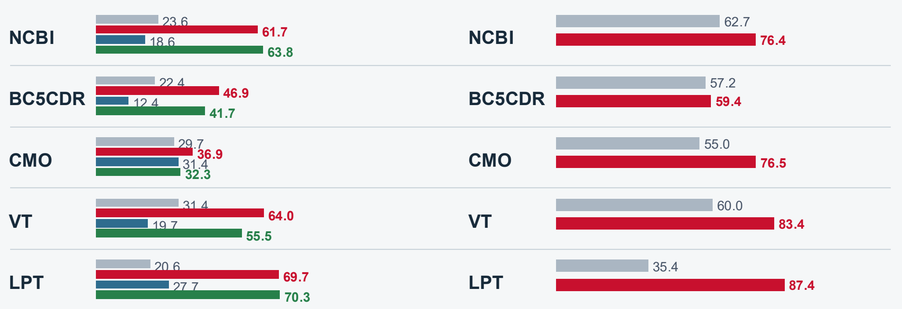


OVERALL RESULTS



Overall: On LPT, BLINK reranker Recall@1 rises from 57.34% to 80.70% (+23.36 points).

NO WORD OVERLAP: THE HARDEST CASES



No overlap: on LPT, BLINK reranker Recall@1 rises from 20.62% to 69.74% (+49.12 points).

Limitations and Future Work

LIMITATIONS

- Alias quality depends on the language model's domain knowledge.
- Evaluation, prompts, and supporting models are English-only.
- Ontology updates require new aliases and filtering.
- The framework does not handle NIL entities: mentions without a valid referent in the target ontology.

FUTURE WORK

- Test multilingual language and similarity models.
- Regenerate aliases when the ontology changes.
- Identify NIL mentions that lack a valid ontology referent.
- Study how LLM domain knowledge affects alias quality.

Related Work

GENERAL DOMAIN

BLINK (Wu et al., 2020) and ReS (Xu et al., 2023b) rank possible matches. Their pretrained representations can miss specialist language.

BIOMEDICAL LINKING

Sung et al. (2020) and Yuan et al. (2022) adapt linking to medical text, but require expert-labeled training examples.

LANGUAGE MODEL AUGMENTATION

Xin et al. (2025) and Ye and Mitchell (2025) generate per mention. Sci-ZSEL generates reusable aliases for selected entities and filters drifted aliases using the ontology hierarchy.

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